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Systems, methods, and mediums for receiving reminders and/or identifying available goods and/or services

Abstract

Examples are generally described that include methods and systems for providing a reminder and/or identifying availability of a good or service, or both. In many examples, a reminder may be provided when a computing device satisfies one or more criteria associated with a reminder request. For instance, a reminder may be provided when the location associated with a computing device is proximate a destination and when the direction of travel associated with a computing device indicates that the proximity of the computing device is increasing or decreasing relative to a predetermined location. In other examples, a computing device may be utilized to identify an entity that associated with a requested good or a requested service, or both. In some examples, a location associated with the computing device may be determined and servers associated with an entity within a range of the location may be searched to identify the request good or the requested service, or both.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6128571	12/1999	Ito	N/A	N/A
6564143	12/2002	Alewine	701/438	G01C 21/26
6601073	12/2002	Robare	N/A	N/A
6631321	12/2002	Ciprian	N/A	N/A
6957076	12/2004	Hunzinger	455/456.3	H04W 4/029
7847686	12/2009	Atkins	340/988	G06Q 10/1095
8244311	12/2011	Boss	455/574	H04W 52/0264
8838146	12/2013	Yoon	455/418	H04W 4/023
10225683	12/2018	Howard	N/A	H04W 4/02
2001/0029410	12/2000	Obradovich	N/A	N/A
2002/0034292	12/2001	Touriniemi	N/A	N/A
2002/0120370	12/2001	Parupudi	N/A	N/A
2002/0138196	12/2001	Polidi	N/A	N/A
2002/0152264	12/2001	Yamasaki	N/A	N/A
2003/0117382	12/2002	Pawlowski	345/204	G09G 5/006
2003/0195819	12/2002	Chen	N/A	N/A
2003/0217160	12/2002	Gayama	N/A	N/A
2003/0224762	12/2002	Lau	455/412.2	H04W 4/02
2003/0225589	12/2002	Eaton	705/14.39	H04L 9/40
2004/0041538	12/2003	Sklovsky	320/127	H04W 52/0264
2004/0204829	12/2003	Endo	N/A	N/A
2004/0230685	12/2003	Seligmann	709/228	G06Q 10/109
2005/0143097	12/2004	Wilson	455/456.3	H04W 64/00
2005/0252965	12/2004	Ohwa	235/383	G06F 16/9537
2005/0273493	12/2004	Buford	709/204	H04L 51/04
2006/0101293	12/2005	Chandley	713/300	G06F 1/3203

2006/0229802	12/2005	Vertelney	N/A	N/A
2007/0005363	12/2006	Cucerzan	704/256	H04W 4/025
2007/0202887	12/2006	Counts	455/456.1	G01S 5/02
2008/0043676	12/2007	Mousseau	370/331	H04L 51/214
2008/0079566	12/2007	Singh	340/539.13	G06Q 10/109
2008/0080682	12/2007	Ogunwale	N/A	N/A
2008/0125103	12/2007	Mock	N/A	N/A
2008/0195312	12/2007	Aaron	455/418	H04L 67/18
2008/0228387	12/2007	Yang	707/999.005	G01C 21/343
2008/0248815	12/2007	Busch	455/456.5	G06Q 30/0255
2009/0006418	12/2008	O'Malley	N/A	N/A
2009/0105947	12/2008	Nachesa	701/533	G01C 21/3679
2009/0157289	12/2008	Graessley	N/A	N/A
2009/0164110	12/2008	Basir	N/A	N/A
2009/0309541	12/2008	Walrath	320/107	G06F 1/3287
2009/0318168	12/2008	Khosravy	N/A	N/A
2010/0216509	12/2009	Riemer	455/557	H04M 1/72463
2010/0250131	12/2009	Relyea	701/469	H04W 4/185
2010/0295676	12/2009	Khachaturov	340/309.7	G08B 21/24
2011/0010220	12/2010	Burnham	705/7.19	H04W 4/18
2011/0057790	12/2010	Martin	340/539.13	H04L 67/306
2011/0131311	12/2010	van Steenberg	709/223	H04L 67/04
2011/0238671	12/2010	Walker	707/E17.084	G06F 16/24578
2014/0180817	12/2013	Zilkha	705/14.55	G06F 16/24575
2016/0360382	12/2015	Gross	N/A	G06F 9/445

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
102009054002	12/2009	DE	G01C 21/3423

OTHER PUBLICATIONS

Merriam-Webster Dictionary, s.v. “criterion,” accessed Nov. 18, 2024 <https://www.merriam-webster.com/dictionary/criterion> (Year: 2024). cited by examiner
http://www.reqall.com/about/remember_here_and_now. cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 12/848,849, filed Aug. 2, 2010, now pending, the disclosure of

which is herein incorporated by reference in its entirety.

BACKGROUND

(1) Unless otherwise indicated herein, the materials described in this section are not admitted to be prior art to the claims by inclusion in this section.

(2) Sometimes it is difficult for one to remember to complete a task at a convenient time for completing the task. For example, when one is near a grocery store and needs to purchase milk, one may forget to stop at the grocery store and purchase milk, or when one is on their way home from work and approaching their dry cleaner, one may forget that they wish to pick up their dry cleaned clothes. In some cases, one may remember to complete a task, but may not be aware of where to go to in order to complete the task, e.g., may not be aware of where to go to purchase a desired good or service. This may be particularly frustrating when one is out shopping and the stores visited do not have the desired good or service.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The foregoing and other features of the present disclosure will become more fully apparent from the following description and appended claims, taken in conjunction with the accompanying drawings. Understanding that these drawings merely depict several examples in accordance with the disclosure, and therefore, are not to be considered limiting of its scope; the disclosure will be described with additional specificity and detail through use of the accompanying drawings.

(2) FIG. 1 illustrates a system for providing a reminder, such as a reminder message, in accordance with at least some examples of the present disclosure.

(3) FIG. 2 illustrates a system for identifying availability of a good or a service, in accordance with at least some examples of the present disclosure.

(4) FIG. 3 is a block diagram illustrating an example system, in accordance with at least some examples of the present disclosure.

(5) FIG. 4 illustrates another system for identifying availability of a good or a service, in accordance with at least some examples of the present disclosure.

(6) FIG. 5 is a flow chart illustrating an exemplary method, in accordance with at least some examples of the present disclosure.

(7) FIG. 6 is a flow chart illustrating another exemplary method, in accordance with at least some examples of the present disclosure.

(8) FIG. 7 is a flow chart illustrating yet another exemplary method, in accordance with at least some examples of the present disclosure.

(9) FIG. 8 is a flow chart illustrating yet another exemplary method, in accordance with at least some examples of the present disclosure.

(10) FIG. 9 is a flow chart illustrating yet another exemplary method, in accordance with at least some examples of the present disclosure.

SUMMARY

(11) The present disclosure describes methods and systems for providing a reminder, such as a reminder message. An exemplary method may include receiving a first location associated with a computing device and a direction of travel associated with the computing device. The method also may include comparing the location associated with the computing device to a first criteria associated with a reminder request and comparing the direction of travel associated with the computing device to a second criteria associated with the reminder request. In some examples, the first criteria may comprise the location associated with the computing device being proximate a destination, and the second criteria may comprise the direction of travel associated with the computing device indicating that the proximity of the computing device to the destination is

increasing. The method also may include providing a reminder associated with the reminder request in response to satisfaction of each criteria associated the reminder request, such as the first criteria and the second criteria.

(12) The present disclosure further describes methods and systems for identifying availability of a good or a service, or both. An exemplary method may include receiving a request for the good or the service, or both, from a computing device. The method further may include receiving a location associated with the computing device. The request for the good or the service, or both, may be provided to at least one server associated with an entity that is located within a particular range of the location associated with the computing device. The method further may include receiving a first message from at least one of the at least one server indicating availability of the requested good or the requested service, or both. A second message indicating the available of the requested good or service may be provided, such as to the computing device.

(13) Another exemplary method may include receiving a request for a good or a service, or both from a computing device. The method further may include selecting an available good or an available service, or both that correlates to the requested good or the requested service, or both. A message may be provided to the computing device indicating the selected available goods or the selected available services, or both, and identifying an entity associated with the available goods or the available services, or both.

(14) An exemplary system may include a receiver, a comparator, and a transmitter. The receiver may be configured to receive a request for a good or a service, or both, from a computing device. The comparator may be configured to compare the requested good or the requested service, or both, to one or more available goods or available services, or both, and to determine whether there is a correlation between the requested goods or the requested services, or both. The transmitter may be configured to provide, in response to the determination that there is a correlation, a message indicating the correlation and identifying at least one entity associated with the correlation.

(15) The foregoing summary is illustrative only and is not intended to be in any way limiting. In addition to the illustrative aspects, embodiments, and features described above, further aspects, embodiments, and features will become apparent by reference to the drawings and the following detailed description.

DETAILED DESCRIPTION

(16) In the following detailed description, reference is made to the accompanying drawings, which form a part hereof. In the drawings, similar symbols typically identify similar components, unless context dictates otherwise. The illustrative examples described in the detailed description, drawings, and claims are not meant to be limiting. Other examples may be utilized, and other changes may be made, without departing from the spirit or scope of the subject matter presented herein. It will be readily understood that the aspects of the present disclosure, as generally described herein, and illustrated in the Figures, can be arranged, substituted, combined, separated, and designed in a wide variety of different configurations, all of which are implicitly contemplated herein.

(17) This disclosure is drawn, inter alia, to methods, systems, devices, computer-readable mediums, and/or apparatus generally related to receiving reminders, such as reminder messages, and identifying available goods and/or services. Some exemplary methods, systems, devices, computer-readable mediums, and/or apparatus may provide a reminder, such as a reminder message, which indicates an availability of a good or a service, or both. In some examples, a reminder may be provided when an attribute, e.g., a location or a direction of travel, or both, associated with a computing device satisfies one or more criteria associated with a reminder request. For instance, a reminder may be provided when the computing device is traveling, e.g., within a moving vehicle, in a particular direction relative to a particular, e.g., predetermined, location and/or is proximate to, e.g., within a predetermined distance of, the particular location. Other exemplary methods, systems, devices, computer-readable mediums, and/or apparatus may include identifying an entity associated

with a requested good or a requested service, or both. In some examples, a location associated with a computing device requesting the good or the service, or both, may be determined and a server associated with an entity within a range of the location may be searched to identify the requested good or the requested service, or both.

(18) Reminders accordingly may be provided when an attribute associated with a computing device satisfies one or more criteria associated with a reminder request. Any of a variety of attributes may be used, including a location or a direction of travel, or both. Moreover, those of ordinary skill in the art readily will understand that the phrase: “direction of travel associated with a computing device,” as used herein, broadly may include a current direction of travel associated with the computing device and/or an anticipated/expected direction of travel, e.g., an anticipated/expected path of travel, associated with the computing device. Those of ordinary skill in the art also readily will understand that the phrase: “location associated with a computing device” broadly may include a current location of the computing device and/or an anticipated/expected future location of the computing device, e.g., a location anticipated/expected based on the anticipated/expected path of travel associated with the computing device. For example, an anticipated or an expected path of travel associated with the computing device may be used to trigger a reminder. The path of travel may, in some examples, be anticipated or expected based on a destination input into a navigation system. In other examples, the path of travel may be anticipated or expected based on past behavior and one or more measured positions or directions. For example, a computing device may identify an expected path of travel from communication with a navigation system or with a device-resident navigation application. The computing device may implement a heuristic that may combine any combination of the current location, the current time, and past travel history. In one example, a phone with a GPS unit may have data related to a path taken on prior trips, e.g., prior trips to work. If the day is a weekday and the time is the morning, an application on the phone may deduce an expected path based on the prior trips to work. In another example, if during prior travel a vehicle associated with the computing device generally or always exits a particular route, e.g., route **405**, via a first exit after entering the particular route via a second exit, the application may deduce that the vehicle associated with the computing device likely will exit the particular route via the first exit after the computing device receives an indication that the vehicle entered the particular route via the second exit.

(19) FIG. **1** illustrates a system **100** for providing a reminder, such as a reminder message, in accordance with at least some examples of the present disclosure. System **100** may include a computing device **110** coupled to a service provider **120** via a network **130**. Computing device **110** may be any type of mobile computing device, such as a laptop computer, a wireless phone, a global positioning system, or any other device capable of transmitting signals to a network.

(20) Network **130** may be any environment configured to receive and to transmit signals. In some examples, the network may be a wide area network (WAN), a local area network (LAN), a campus area network (CAN), a metropolitan area network (MET), a personal area network (PAN), or the like.

(21) Computing device **110** may be configured to allow a user to generate one or more reminder requests. A reminder request may include or correspond to a reminder, such as a reminder message. The reminder may remind a user of something, such as a task to complete. A reminder request may include one or more criteria that, when satisfied, trigger the reminder to be provided to computing device **110**. In some examples, the one or more criteria may include a class of stores, such as a drugstore or a grocery store, a class of goods, such as books, or a class of services, such as dry cleaning or car washing, or any combination thereof. The one or more criteria also may include a trademark, a good brand name, a store brand name, or a service brand name, or any combination thereof. In some examples, the one or more criteria may include one or more particular, e.g., predetermined, destinations or locations. In some examples, the reminder request may include a criteria that computing device **110** be within a predetermined distance of a first location to trigger

the reminder to be provided to computing device **110**. The one or more criteria also may include criteria associated with a direction of travel associated with computing device **110**, such as the direction of travel associated with computing device **110** indicating that the proximity of computing device **110** to the first location and/or second location, such as a user's home or office, is increasing. In some examples, the one or more criteria further may include a time of day, such as after work or during daylight hours. Other time-related criteria may include a calendar event, such as payday, wife's birthday, every second Tuesday, etc. Criteria may include factors other than location or destination information in some examples. For example, a criteria may be when the computing device is connected to a network having a predetermined speed or cost. In one example, a reminder may be provided, for example, to download an audio or video file when the computing device is connected to a free network or network having below a predetermined cost threshold per bit.

(22) The reminder may be a visual reminder and/or an audio reminder, and may include a task to be completed. In some examples, the reminder may include information indicating how the one or more criteria of the reminder request were satisfied and/or directions to the particular location associated with the satisfaction of the one or more criteria. For instance, if a reminder request includes criteria associated with computing device **110** being within a predetermined number of blocks, e.g., two blocks, of a particular class of stores, e.g., grocery stores, and a direction of travel associated with computing device **110** corresponding to a direction of travel indicating that computing device **110** is moving closer to the particular class of stores, the reminder may provide directions to the grocery store and remind the user to purchase a particular item, such as milk.

(23) Computing device **110**, network **130**, or service provider **120**, or any combination thereof, may be configured to determine a location and/or a direction of travel associated with computing device **110**, such as an approximate location of computing device **110** or a range in which computing device **110** is located. For instance, computing device **110** may be configured to determine a location and/or a direction of travel associated with computing device **110** via a positioning means, such as a global positioning system (GPS). In other examples, service provider **120**, computing device **110** and/or network **130**, in combination or by themselves, may be configured to determine a location and/or a direction of travel associated with computing device **110** using cell tower signal strength and/or a cell tower triangulation. In particular, distance to a cell tower may be determined by signal strength to one or more cell towers to associate computing device **110** with a location. As signal strength to one or more towers increases or decreases, a direction of travel may be identified. In one example, network **130** may provide signal strength information and a cell tower location to service provider **120**, and service provider **120** may analyze the information to determine a location and/or a direction of travel associated with computing device **110**. In one example, network **130**, service provider **120**, or computing device **110**, or any combination thereof, may be configured to detect a speed associated with the movement of computing device **110** using locations associated with computing device **110** over time. In some examples, the presence of a signal itself may provide location information. For example, an SSID received from a known local area network, such as a home or office network, may itself provide location information (e.g., home or work location).

(24) In one example, computing device **110** may be configured to generate and to store reminder requests and to trigger at least one reminder request when at least one attribute associated with computing device **110** satisfies one or more criteria associated with the reminder request. In some examples, computing device **110** may comprise software configured to compare the location and/or the direction of travel associated with computing device **110** to the one or more criteria associated with the one or more reminder requests. In this example, computing device **110** may be coupled to network **130**, but may not be coupled to service provider **120** via network **130**. Computing device **110** may be configured to determine whether at least one attribute associated with computing device **110** satisfies one or more criteria associated with the reminder request. For example, computing device **110** may be configured to compare a location associated with computing device

110 with a first criteria associated with a first reminder request to determine whether the first criteria is satisfied. The first criteria may be satisfied when computing device **110** is proximate a predetermined destination, e.g., within a predetermined distance of the predetermined destination, such as a particular store. The term: “within” is one example of a logical qualifier that may be evaluated by the computing device **110**. Other logical qualifiers may include NOT, OR, AND, or any combination thereof. For example, one criteria may be that it is not Saturday. Another criteria may be based on proximity to a first store or a second store. Yet another criteria may be when it is the afternoon and it is cloudy. Computing device **110** may be configured to compare a direction of travel associated with computing device **110** with a second criteria associated with the first reminder request to determine whether the second criteria is satisfied. The second criteria may be satisfied when the direction of travel associated with computing device **110** indicates that the proximity of computing device **110** to the predetermined destination is increasing. In response to each of the one or more criteria being satisfied, computing device further may be configured to provide a reminder, e.g., a reminder message, associated with the reminder request. In one example, computing device **110** may be configured to generate the reminder and to provide the reminder to an output of computing device **110**. In other examples, computing device **110** may be configured to provide the reminder to another computing device.

(25) In another example, service provider **120** may be configured to receive and to store reminder requests and to trigger at least one reminder request when at least one attribute associated with computing device **110** satisfies one or more criteria associated with the reminder request. Service provider **120** may be configured to receive, via network **130**, the one or more reminder requests from computing device **110**. When at least one attribute associated with computing device **110** satisfies at least one or all criteria associated with the reminder request, service provider **120** may be configured to provide the reminder corresponding to the reminder request to computing device **110** or to cause a reminder stored on computing device **110** to be provided to an output of computing device **110**.

(26) In an another example, service provider **120** or computing device **110** may be configured to send a corresponding reminder to a second computing device (not shown) when an attribute associated with computing device **110** satisfies the one or more criteria associated with a reminder request. The reminder may be sent to the second computing device without a reminder being sent to computing device **110**. In another example, the reminder sent to the second computing device may be the same reminder sent to an output of computing device **110** or may be a different reminder. For instance, if a first user of computing device **110** is near a grocery store and receives a first reminder including directions to the grocery store and a message reminding the user to pick up milk, a second user of a second computing device may receive a second reminder indicating that computing device **110** is near the grocery store.

(27) Computing device **110** may be configured to prevent the generation of reminders even when attributes associated with computing device **110** satisfy each criteria associated with a reminder request. For example, computing device **110** may comprise a feature that allows a user to prevent the triggering of a first reminder but allow the triggering of a second reminder even though computing device **110** satisfies each criteria associated with both the first and second reminders. In some examples, when computing device **110** is traveling at a relatively constant speed, such as when on a plane or a train, computing device **110** may be configured to prevent the generation of a reminder even when attributes associated with computing device **110** satisfies each criteria associated with the reminder.

(28) FIG. 2 illustrates a system **200** for identifying an availability of a good or a service, or both, in accordance with at least some examples of the present disclosure. System **200** may include a computing device **210** coupled to a service provider **220** via a network **230**. Service provider **220** may be further coupled to one or more entity server **240** via network **230**. Computing device **210** and network **230** may be substantially similar to computing device **110** and network **130**,

respectively. Moreover, each entity server **240** may be associated with an entity, such as a provider of goods, e.g., a store, a service provider, or any other type of entity.

(29) A user of computing device **210** may transmit a request for a good or a service, or both, to service provider **220** via network **230**. The term: “user,” as used herein, may include designates or software applications, and is not limited to human individuals. In some examples, an individual may interact with an application on computing device **210** that may have rules for when to transmit and/or synchronize with a server or service provider **220**. Service provider **220** may be configured to receive and to store the request for the good or the service, or both. The request for a good or a service may include a company brand name, a trademark, a good type, a good brand, a good name, a service type, a service brand, a service name, a sku, a barcode scan, or any combination thereof, and may include any other type of information describing a supplier, a good, and/or a service.

(30) Service provider **220** may be configured to determine whether an entity, e.g., an entity located within a particular range of a location associated with computing device **210**, has the requested good or the requested service, or both, available, and satisfies each criteria associated with the request. The criteria for the request may include any criteria described with respect to FIG. **1**. In one example, service provider **220** may be configured to receive information, such as a good or a service list, from one or more entity server **240**. The information or lists received from entity server **240** may include one or more available good or one or more available service associated with the entity. The available goods or available services may include a company brand name, a trademark, a good type, a good brand, a good name, a service type, a service brand, a service name, or any combination thereof, and may include any other type of information describing a supplier, good, or service. Service provider **220** may be configured to compare the available goods or the available services, or both, with the requested good or the requested service, or both, to identify a correlation therebetween. In response to identifying the correlation, service provider **220** may be configured to generate a message indicating the correlation. For example, such a message may indicate the availability of the requested good or the requested service, or both, and information identifying the entity associated with the available good or the available service, or both. The information or lists received from entity server **240** may be maintained by service provider **220** and may be regularly updated by entity server **240** via network **230**, such daily or weekly.

(31) In another example, service provider **220** may be configured to provide the requested good or the requested service to an entity server **240** associated with an entity that is located within the particular range of the location associated with computing device **110**. Service provider **220** also may be configured to receive a message from the entity server **240** associated with the entity indicating the availability of the requested good or the requested service, or both. In response to receiving the message indicating the availability of the requested good or the requested service, or both, service provider **220** may be configured to generate a message indicating the availability of the requested good or the requested service, or both, and information identifying the entity associated with the available good or the available service, or both. In any of the above-described examples, service provider **220** may be configured to provide the message to computing device **210** via network **230**. The message may indicate the availability of the good or the service, or both, and may include a name, an address, or directions to the entity associated with the available good or service, or any combination thereof. In some examples, the message may include a map illustrating computing device's **210** location and the location of the entity associated with the available good or the available service, or both.

(32) In some examples, service provider **220** may include logic configured to adjust the location range associated with the request for an available good or service. For instance, service provider **220** may be configured to compare the requested goods or the requested services to available goods or available services associated with entities located outside of the range included in the request. In another example, the location range may be increased based on the request indicating a level of importance associated with the requested good or the requested service. In one example, service

provider is configured to utilize the location and the direction of travel associated with the computing device **210**, which may include data from a navigation system, a navigation application, an event calendar, an expectation based on past history, or any combination thereof, to identify areas that computing device **210** is expected to travel and to communicate with entity servers **240** for entities in the identified areas.

(33) In another example, service provider **220** is configured to review consumer ratings for each entity identified as having an available good or an available service, or both, that correlates to the requested good or the requested service. Service provider **220** may be configured to provide the message indicating a name associated with the entity, an address associated with the entity, directions thereto, or any combination thereof, in the event the consumer ratings for the entity are above a particular threshold.

(34) FIG. **3** is a block diagram illustrating an example system **300**, in accordance with at least some examples of the present disclosure. System **300** may include a processor **305**, a comparator **310**, a selector **320**, a storage medium **330**, and a positioning locator **360**. In the example illustrated in FIG. **3**, comparator **310**, selector **320**, storage medium **330**, and positioning locator **360** are on processor **305**. However, any or all of these components may be separate from processor **305**. Processor **305** may be coupled to a transmitter **340** and a receiver **350** to receive and to transmit signals via network **130**. Storage medium **330** may be configured to store the reminders and/or the requested good or the requested service, or both, received from computing device **110**. Storage medium **330** also may be configured to store good or service lists received from entity servers **240**. Comparator **310** may be configured to compare the requested good or the requested service, or both, to the available goods or the available services indicated on the store good or service lists. Comparator **310** further may be configured to compare the one or more criteria in a reminder request to a location and/or a direction of travel associated with computing device **110**, such as the direction of travel relative to a predetermined location. Selector **320** may be configured to select one of the reminder requests to be sent to computing device **110** in response to more criteria being satisfied. Selector **320** further may be configured to select one of the entities in response to the good or service list including an available good or an available service, or both, that correlates with the requested good or service. Transmitter **340** may be configured to provide, in response to the determination that there is a correlation, a message, indicating the correlation and identifying at least one entity associated with the correlation. Positioning location **360** may be configured to identify a location, such as a range of location or an approximate location, associated with computing device **110**, such as by the systems and methods discussed above.

(35) In some examples, storage medium **330** may store program instructions that when implemented by processor **305** cause processor **305** to compare the requested good or the requested service, or both, to the available goods or available services, or both, indicated on the store good or service lists, to compare the one or more criteria in a reminder request to a location and a direction of travel associated with computing device **110**, and to select one of the reminder requests to be sent to computing device **110** in the event of one or more criteria in a reminder request being satisfied. Service providers **120** and **220** and/or computing device **110** may include system **300**.

(36) FIG. **4** illustrates another system **400** for identifying availability of a good or service, in accordance with at least some examples of the present disclosure. System **400** may be substantially similar to system **200**, except that computing device **210** is coupled to the one or more entity servers **240** via an ad hoc network **150**. In one example, computing device **210** may be configured to automatically connect to entity servers **240** when computing device **210** comes into range of an ad hoc network **150**. In another example, computing device **210** may have security controls implemented for connection to entity servers **240**. For example, in one embodiment, only request/response pairs for the availability service may be allowed access to ad hoc network **150**. In other examples, computing device **210** may be configured to request permission from a user of computing device **210** to connect to the ad hoc network **150**. Upon computing device **210** being

coupled to entity server **240**, computing device **210** may upload one or more requested goods or requested services, or both, or alternatively, entity server **240** may download to computing device **210** a list of available goods or available services, or both. In the former case, entity server **240** may compare the requested good or requested, or both service with the available goods or the available services, or both, corresponding to one or more entities. In the latter case, computing device **210** may compare the requested good or the requested service, or both, to the available goods or the available services, or both, on the list. In response to a correlation between the available goods or the available services, or both, and the requested goods or the requested services, or both, a message indicating the correlation may be provided to the user of computing device **210**. In some examples, requested goods or services may be associated with a priority, and reminders for low-priority items may be suppressed when a power level of computing device **210** is below a predetermined power level or computing device **210** is operating in a power conservation mode. This may avoid the initiation of network traffic that may reduce available battery time of computing device **210**. The priority associated with the requested goods or services may be user input or automatically may be determined.

(37) In this example system **400**, computing device **210** may not provide a location associated with computing device **210** to the entity server **240**. Rather, the fact that computing device **210** is sufficiently close to ad hoc network **150** indicates that computing device **210** is within a range of entity server **240**.

(38) In one example, the entity that sells or offers the available good or the available service may be the same entity that maintains the entity server **240**. In another example, entity server **240** may be maintained by one or more entities within a relatively close range, such as many stores in a mall.

(39) In some examples, computing device **210** may include a feature that temporarily prevents service providers **120** and **220** from sending messages even in the event of a correlation between the requested good or the requested service, or both, and the available goods or the available services, or both.

(40) FIG. 5 is a flow chart illustrating an exemplary method **500**, in accordance with at least some examples of the present disclosure. Method **500** may include one or more functions, operations, or actions, as illustrated by blocks **510-560**, for a computing device to provide a message. Method **500** may begin at block **510**. In block **510**, a first location associated with the computing device may be received. Block **510** may be followed by block **520**. In block **520**, a direction of travel associated with the computing device may be received. Block **520** may be followed by block **530**. In block **530**, the location associated with the computing device may be compared to a first criteria associated with a reminder request. Block **530** may be followed by block **540**. In block **540**, the direction of travel associated with the computing device may be compared to a second criteria associated with the reminder request. Block **540** may be followed by block **550**. In block **550**, a reminder message associated with the selected request may be provided in response to satisfaction of each criteria associated with the reminder request.

(41) FIG. 6 is a flow chart illustrating another exemplary method **600**, in accordance with at least some examples of the present disclosure. Method **600** may include one or more functions, operations, or actions, as illustrated by blocks **610-650** for identifying availability of a good or a service, or both. Method **600** may begin at block **610**. In block **610**, a request for the good or the service, or both, may be received from a computing device. Block **610** may be followed by block **620**. At block **620**, a location associated with the computing device may be determined. Block **620** may be followed by block **630**. In block **630**, the request for the good or the service, or both, may be provided to at least one server associated with at least one entity that is located within a particular range of the location associated with the computing device. Block **630** may be followed by **640**. In block **640**, a first message indicating availability of the requested good or the request service, or both, may be received. Block **640** may be followed by **650**. In block **650**, a second message indicating the availability of the requested good or the requested service, or both, may be

provided to the computing device.

(42) FIG. 7 is a flow chart **700** illustrating yet another exemplary method **700**, in accordance with at least some examples of the present disclosure. Method **700** may include one or more functions, operations, or actions, as illustrated by blocks **710-740** for identifying availability of a good or service. Method **700** may begin at block **710**. In block **710**, a request for the good or the service, or both, may be received from a computing device. Block **710** may be followed by block **720**. In block **720**, a location associated with the computing device may be determined. Block **720** may be followed by block **730**. In block **730**, the requested good or the requested service, or both, may be compared to available goods or available services from at least one entity located within a particular range of a location associated with the computing device. Block **730** may be followed by block **740**. In block **740**, a correlation between the requested good or the requested service, or both, and the available goods or the available service, or both, may be determined. Block **740** may be followed by block **750**. In block **750**, a message indicating the correlation and identifying at least one entity associated with the correlation may be provided.

(43) FIG. 8 is a flow chart illustrating another exemplary method **800**, in accordance with at least some examples of the present disclosure. Method **800** may include one or more functions, operations, or actions, as illustrated by blocks **810-830** for identifying availability of a good or a service. Method **800** may begin at block **810**. In block **810**, a request for the good or the service may be received from a computing device. Block **810** may be followed by block **820**. At block **820**, the requested good or the requested service, or both may be compared to available goods or available services, or both, Block **820** may be followed by **830**. In block **830**, a correlation between the requested good or the requested service, or both, and the available goods or the available service, or both may be determined. Block **830** may be followed by **840**. In block **840**, a message may be provided to the computing device indicating the correlation and identifying at least one entity associated with the correlation.

(44) FIG. 9 is a flow chart illustrating another exemplary method **900**, in accordance with at least some examples of the present disclosure. Method **900** may include one or more functions, operations, or actions, as illustrated by blocks **910-930** for identifying availability of a good or a service, or both. Method **900** may begin at block **910**. In block **910**, a request for the good or the service, or both, may be received from a computing device. Block **910** may be followed by block **920**. At block **920**, an available good or an available service, or both, that correlates to the requested good or the requested service may be selected. Block **920** may be followed by **930**. In block **930**, a message may be provided to the computing device indicating the selected available good or the selected service, or both.

(45) The various blocks described herein for methods **500**, **600**, **700**, **800**, and **900** may be performed sequentially, in parallel, or in a different order than those described herein. It also should be appreciated that in some implementations one or more of the illustrated blocks may be eliminated, combined, or separated into additional blocks. The described and illustrated methods **500**, **600**, **700**, **800**, and **900** also may include various additional blocks not shown. For instance, in method **500**, a first location associated with the computing device may be received at the same time as a direction of travel associated with the computing device is received.

(46) While various examples have been disclosed herein, other examples will be apparent to those skilled in the art. The various examples disclosed herein are for purposes of illustration and are not intended to be limiting, with the true scope and spirit being indicated by the following claims.

Claims

1. An electronic memory storing instructions, which upon execution by a processing device, causes the processing device to perform operations comprising: compute a geographic location of a computing device in a vehicle, wherein the computing device is configured to allow a person to

input information including a particular good or particular service for generation of a reminder request associated with criteria, wherein the criteria when satisfied by an attribute associated with the computing device including an actual location of the computing device or an actual direction of travel of the computing device, triggers one or more reminders including content about the particular good or the particular service; identify, based at least on the computed geographic location, one or more paths taken by the computing device; following computation of the geographic location of the computing device, provide information about the one or more paths to an application configured to derive a path of travel based at least in part on a history of prior trips from a given geographic location, wherein the application comprises instructions executable by the processing device or a remote processing device, wherein the one or more reminders are based at least in part on the at least one path suggestion and the criteria, and said provide the information about the one or more paths to the application causes the application to derive at least one path suggestion based at least in part on the provided information; generate a visualization of a reminder of the one or more reminders, the reminder including first content including availability of the particular good or the particular service; determine whether to suppress notification, or not, based at least in part on a priority associated with the particular good or the particular service; in a case of a determination to not suppress the notification, display the visualization of the reminder.

2. The electronic memory of claim 1, wherein the at least one path suggestion is derived based at least in part on a day of a week or a time of the day.

3. The electronic memory of claim 1, wherein the criteria associated with the reminder request comprises: an identification of a class or category of stores, a class or category of services, a trademark, a brand name, a store brand name, a service brand name, a predetermined destination, a predetermined location, a predetermined distance from a location, or a time of day, availability of network connection, or a combination thereof.

4. The electronic memory of claim 1, wherein the one or more reminders comprise a plurality of reminders and the reminder comprises a first reminder of the plurality of reminders, and wherein the operations further comprise: transmit a second reminder of the plurality of reminders to a device located remotely from the vehicle, wherein the first and second reminders are triggered by satisfying the criteria, the second reminder including second content that is different than the first content, wherein the second reminder specifies a suggested destination corresponding to the at least one path suggestion.

5. The electronic memory of claim 1, wherein at least a portion of the criteria is included in the reminder request.

6. A system, the system comprising: a receiver configured to receive a signal associated with a location of a computing device and a direction of travel associated with the computing device; a processor; and a memory operationally coupled to the processor, wherein the memory stores program instructions implemented by the processor for: comparing the location associated with the computing device to a first criteria associated with a reminder request, wherein the reminder request is based at least in part on input information including a particular good or a particular service; comparing the direction of travel associated with the computing device to a second criteria associated with the reminder request, wherein the first criteria comprises the location associated with the computing device being proximate a destination, and the second criteria comprises the direction of travel associated with the computing device indicating that the proximity of the computing device to the destination is increasing or is expected to increase; determining whether to suppress notification, or not, based at least in part on a priority associated with the particular good or the particular service; and in a case of a determination to not suppress the notification, providing a reminder associated with the reminder request in response to satisfaction of the first criteria and the second criteria.

7. The system of claim 6, wherein the location associated with the computing device being proximate the destination comprises the location associated with the computing device being

within a predetermined distance of the destination.

8. An electronic memory storing instructions, which upon execution by a processing device, causes the processing device to perform operations comprising: compute a geographic location of a computing device in a vehicle, wherein the computing device is configured to allow a person to input information including a particular good or particular service for generation of a reminder request associated with criteria, wherein the criteria when satisfied by an attribute associated with the computing device including an actual location of the computing device or an actual direction of travel of the computing device, triggers one or more reminders including content about the particular good or the particular service; following computation of the geographic location of the computing device, identify at least one path suggestion, wherein the one or more reminders are based at least in part on the at least one path suggestion and the criteria; generate a visualization of a reminder of the one or more reminders, the reminder including first content including availability of the particular good or the particular service; determine whether to suppress notification, or not, based at least in part on a priority associated with the particular good or the particular service; in a case of a determination to not suppress the notification, display the visualization of the reminder.

9. The electronic memory of claim 8, wherein the operations further comprise: identify, based at least on the computed geographic location, one or more paths taken by the computing device; wherein the at least one path suggestion is based at least in part on information about the one or more paths.

10. The memory of claim 9, wherein the operations further comprise: following computation of the geographic location of the computing device, provide the information about the one or more paths to an application configured to derive a path of travel based at least in part on a history of prior trips from a given geographic location, wherein the application comprises instructions executable by the processing device or a remote processing device, wherein said provide the information about the one or more paths to the application causes the application to derive the at least one path suggestion based at least in part on the provided information.

11. The electronic memory of claim 10, wherein the at least one path suggestion is derived based at least in part on a day of a week or a time of the day.

12. The electronic memory of claim 8, wherein the criteria associated with the reminder request comprises: an identification of a class or category of stores, a class or category of services, a trademark, a brand name, a store brand name, a service brand name, a predetermined destination, a predetermined location, a predetermined distance from a location, or a time of day, availability of network connection, or a combination thereof.

13. The electronic memory of claim 8, wherein the one or more reminders comprise a plurality of reminders and the reminder comprises a first reminder of the plurality of reminders, and wherein the operations further comprise: transmit a second reminder of the plurality of reminders to a device located remotely from the vehicle, wherein the first and second reminders are triggered by satisfying the criteria, the second reminder including second content that is different than the first content, wherein the second reminder specifies a suggested destination corresponding to the at least one path suggestion.

14. The electronic memory of claim 8, wherein at least a portion of the criteria is included in the reminder request.
